

Today's Topic: Diode AC Analysis.

Why is it done?

→ Mostly for Rectification

(Converting AC → DC)

How to perform AC Analysis for a Diode?

Steps:

Step 1: Determine the transition state voltage.

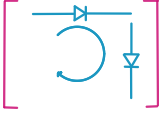
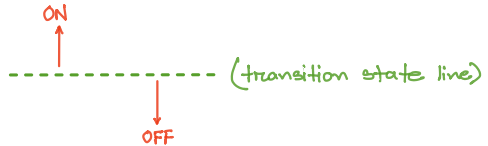
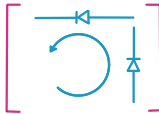
[At transition, $V_D = 0.7V$; $I_D = 0A$]

- Redraw the ckt. assuming the diode is going through transition.
- Apply KVL to determine v_i at which transition occurs.

[Since v_i is unknown, consider it to be +ve.]

Step 2: Draw the transition state voltage line on the v_i (input) curve.

Step 3: Determine the range of v_i for which diode is "ON" and "OFF".

- If the diode orientation is **clockwise**  then  (transition state line)
- If the diode orientation is **anticlockwise**  then  (transition state line)

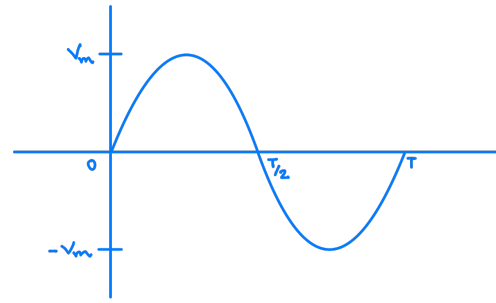
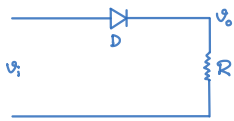
Step 4: Determine the input/output relationship.

- When diode is "ON": Replace with 0.7 and KVL.
- When diode is "OFF": Open ckt and KVL.

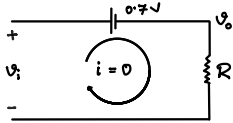
Step 5: Sketch v_o curve based on the input/output relationship.

Let's follow these steps to solve a problem.

Example 1: Find the output waveform.



Solution: At transition, $V_D = 0.7V$ & $i = 0A$



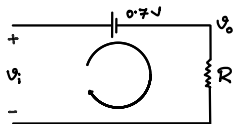
Applying KVL: $v_i - 0.7 - iR = 0 \therefore v_i = 0.7V$

[Here, the diode orientation is clockwise]

$\therefore$ For $v_i > 0.7$: ON $[t_1 < t < t_2]$

$\therefore$ For $v_i \leq 0.7$: OFF $[0 < t < t_1 \text{ \& } t_2 < t < T]$

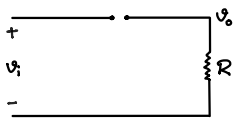
When diode is "ON":



Applying KVL: $v_i - 0.7 - v_o = 0$

$$\therefore v_o = v_i - 0.7$$

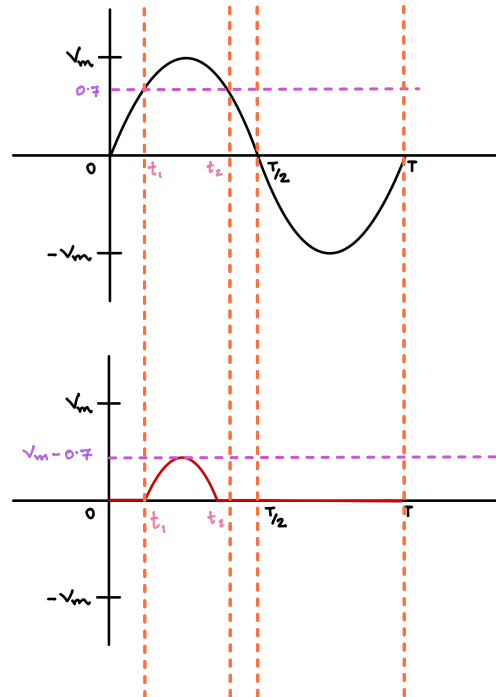
When diode is "OFF":



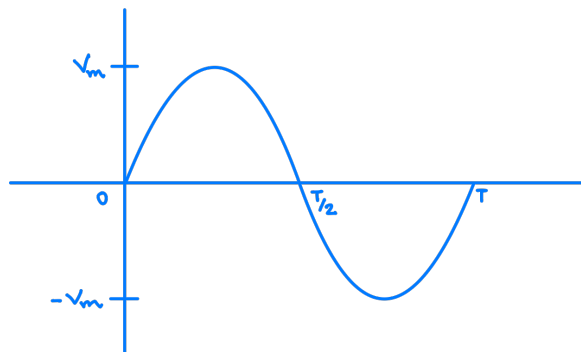
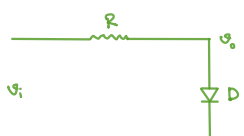
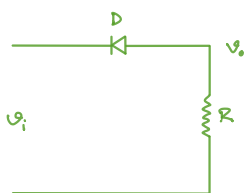
$$\therefore v_o = 0$$

For upper limit, $v_i = V_m$

$$\therefore v_o = V_m - 0.7 = \boxed{V_m - 0.7}$$

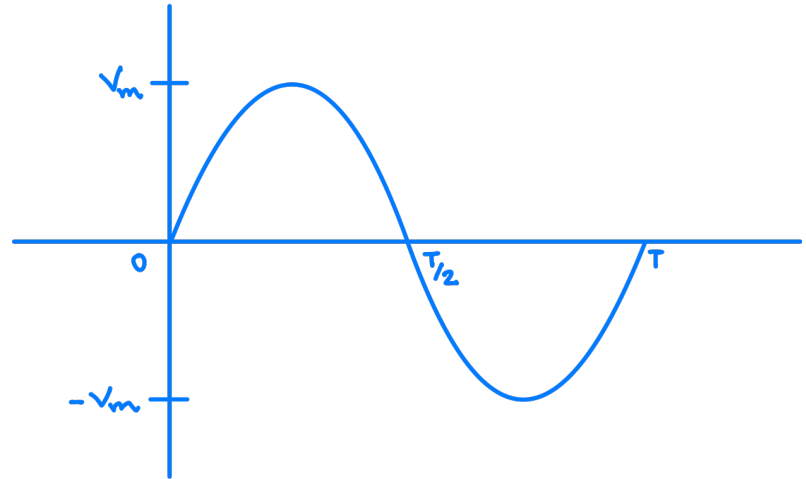
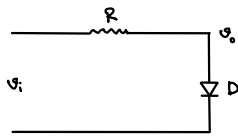


Try to solve yourself:

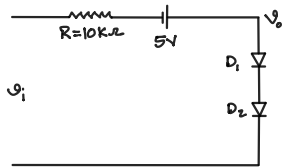


Diode AC Analysis Problems can be of several types (Excluding Clamping and Rectifiers)

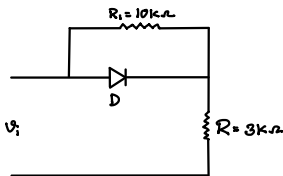
1) Diode in series with resistors:



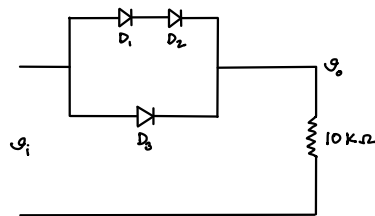
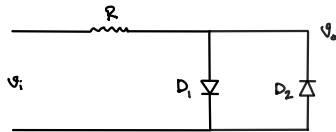
2) Clipper circuits:



3) Resistors in parallel with diodes:



4) Diodes parallel to each other:



5) Diode and DC voltage in series and parallel to a resistor:

